

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q18-1016

Commonwealth Chesapeake 138 kV  
March 03, 2020

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1016. The Interconnection Customer proposes a 69.1 MW generating facility interconnecting at the Commonwealth Chesapeake 138 kV substation in the Virginia area. The request was received on June 24, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 319 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,666,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

For the loss of Southtown – Commonwealth Chesapeake 345 kV, the Quince Orchard – Agawam 138 kV Line loads to 143.1% of its 171 MVA rating. Project DFAX: 9.6%.

Commonwealth Chesapeake – Rodessa 138 kV: 140.6% loading (232 MVA) following the outage of the Southtown – Commonwealth Chesapeake Substation 345 kV; DFAX = 12.4%.

The Southtown 345/138kV Auto is overloaded to 133.7% of its 1,078 MVA emergency rating for the loss of the Southtown – Vienna Operations 345 kV. The project contributes 38.2% DFAX (26.4 MW) to this overload.

Southtown – Quince Orchard 138 kV: 132.9% loading (336 MVA) following the outage of the Commonwealth Chesapeake – Southtown 345 kV; DFAX = 19.3%.

For the loss of Huntington Falls – Otay Lake 138 kV, the Southtown – Commonwealth Chesapeake 345 kV L loads to 131.8% of its 1,871 MVA rating. Project DFAX: 72.8%.

The Shingle Crk – West -Station 138 kV L is overloaded to 131.4% of its 906 MVA emergency rating for the loss of the Vienna Operations – Southtown 345 kV. The project contributes 53.7% DFAX (37.1 MW) to this overload.

For the loss of Eston 2 – Cady 345 kV, the Wye Mills – Grasonville 138 kV loads to 129.9% of its 203 MVA rating. Project DFAX: 6.4%.

For the loss of Vienna Operations – Southtown 345 kV, the Southtown – Shingle Crk 138 kV loads to 129.9% of its 940 MVA rating. Project DFAX: 53.7%.

Quince Orchard – Colchester 138 kV: 129.2% loading (556 MVA) following the outage of the Commonwealth Chesapeake – Southtown 345 kV L; DFAX = 28.9%.

For the loss of Vienna Operations – Easton 2 345 kV, the Vienna Operations 345/1 kV XFMR loads to 128.1% of its 981 MVA rating. Project DFAX: 47.6%.

The Easton 2 – Vienna Operations 345 kV line is overloaded to 127.8% of its 1,710 MVA emergency rating for the loss of the Linde – Ripple 138 kV Line. The project contributes 76.9% DFAX (53.1 MW) to this overload.

The Memphis Jct – Persimmon Gap 138 kV is overloaded to 127.3% of its 409 MVA emergency rating for the loss of the Vienna Operations – Easton 2 345 kV. The project contributes 20.2% DFAX (14.0 MW) to this overload.

For the loss of Southtown – Vienna Operations Substation 345 kV, the Ripple 345 kV/138 kV Transformer loads to 127.1% of its 620 MVA rating. Project DFAX: 35.9%.

Valleyview – Colchester 138 kV L: 127.0% loading (587 MVA) following the outage of the Southtown – Commonwealth Chesapeake 345 kV; DFAX = 28.9%.

The Huntington Falls – Commonwealth Chesapeake 138 kV Line is overloaded to 126.4% of its 748 MVA emergency rating for the loss of the Commonwealth Chesapeake – Southtown 345 kV Line. The project contributes 34.6% DFAX (23.9 MW) to this overload.

For the loss of Southtown – Commonwealth Chesapeake 345 kV Line, the Dane – Prunedale 138 kV L loads to 126.3% of its 303 MVA rating. Project DFAX: 14.3%.

For the loss of Vienna Operations – Easton 2 Substation 345 kV Line, the Saddlebrooke Ranch – Sunset 138 kV loads to 125.7% of its 925 MVA rating. Project DFAX: 42.4%.

Vedder – Otay Lake 138 kV: 125.4% loading (1,040 MVA) following the outage of the Commonwealth Chesapeake – Southtown 345 kV; DFAX = 46.9%.

Huntington Falls – Otay Lake 138 kV: 125.2% loading (1,048 MVA) following the outage of the Southtown – Commonwealth Chesapeake 345 kV; DFAX = 46.9%.

Tulucay – Michoud Sta 138 kV: 124.8% loading (167 MVA) following the outage of the Southtown – Commonwealth Chesapeake 345 kV line; DFAX = 8.4%.

Hillsboro – Saltspring Island 138 kV: 124.3% loading (242 MVA) following the outage of the Cady – Easton 2 345 kV L; DFAX = 11.7%.

The Quinn – Bulverde 138 kV is overloaded to 123.6% of its 406 MVA emergency rating for the loss of the Vienna Operations – Easton 2 345 kV. The project contributes 16.9% DFAX (11.7 MW) to this overload.

For the loss of Southtown – Commonwealth Chesapeake 345 kV, the Langlois – Southtown 138 kV loads to 122.2% of its 520 MVA rating. Project DFAX: 27.4%.

The Dane – Commonwealth Chesapeake 138 kV is overloaded to 122.0% of its 664 MVA emergency rating for the loss of the Commonwealth Chesapeake Substation – Southtown 345 kV. The project contributes 29.0% DFAX (20.0 MW) to this overload.

The Key Largo – Saltspring Island 138 kV is overloaded to 121.6% of its 243 MVA emergency rating for the loss of the Easton 2 – Cady 345 kV Line. The project contributes 11.7% DFAX (8.1 MW) to this overload.

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                     | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Quince Orchard – Agawam 138 kV Line          | Rebuild line with higher-capacity conductor | \$3,312,000   | 3% (\$89,000)         |
| Commonwealth Chesapeake – Rodessa 138 kV     | Rebuild line with higher-capacity conductor | \$4,249,000   | 2% (\$105,000)        |
| Southtown 345/138 kV Transformer             | Replace transformer with higher-rated unit  | \$15,134,000  | 2% (\$262,000)        |
| Quince Orchard – Southtown 138 kV Line       | Rebuild line with higher-capacity conductor | \$2,683,000   | 3% (\$70,000)         |
| Southtown – Commonwealth Chesapeake 345 kV L | Rebuild line with higher-capacity conductor | \$64,680,000  | 2% (\$1,337,000)      |

| Facility                                          | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| West -Station – Shingle Crk 138 kV                | Rebuild line with higher-capacity conductor | \$14,722,000  | 2% (\$324,000)        |
| Grasonville – Wy Mills 138 kV Line                | Rebuild line with higher-capacity conductor | \$16,776,000  | 2% (\$301,000)        |
| Southtown – Shingle Crk 138 kV                    | Rebuild line with higher-capacity conductor | \$6,241,000   | 2% (\$137,000)        |
| Quince Orchard – Colchester 138 kV                | Rebuild line with higher-capacity conductor | \$2,432,000   | 3% (\$63,000)         |
| Vienna Operations 345/1 kV XFMR                   | Replace transformer with higher-rated unit  | \$10,709,000  | 2% (\$234,000)        |
| Easton 2 – Vienna Operations 345 kV line          | Rebuild line with higher-capacity conductor | \$91,678,000  | 2% (\$1,853,000)      |
| Persimmon Gap – Memphis Jct 138 kV                | Rebuild line with higher-capacity conductor | \$3,180,000   | 2% (\$60,000)         |
| Ripple 345 kV/138 kV Transformer                  | Replace transformer with higher-rated unit  | \$15,198,000  | 2% (\$341,000)        |
| Colchester – Valleyview 138 kV                    | Rebuild line with higher-capacity conductor | \$25,306,000  | 3% (\$660,000)        |
| Commonwealth Chesapeake – Huntington Falls 138 kV | Rebuild line with higher-capacity conductor | \$4,285,000   | 3% (\$113,000)        |
| Dane – Prunedale Substation 138 kV                | Rebuild line with higher-capacity conductor | \$3,624,000   | 3% (\$95,000)         |
| Saddlebrooke Ranch – Sunset 138 kV                | Rebuild line with higher-capacity conductor | \$14,489,000  | 2% (\$282,000)        |
| Vedder – Otay Lake 138 kV                         | Rebuild line with higher-capacity conductor | \$6,196,000   | 3% (\$160,000)        |
| Huntington Falls – Otay Lake 138 kV               | Rebuild line with higher-capacity conductor | \$8,304,000   | 3% (\$215,000)        |
| Tulucay – Michoud Sta 138 kV                      | Rebuild line with higher-capacity conductor | \$30,944,000  | 3% (\$830,000)        |
| Hillsboro – Saltspring Island 138 kV              | Rebuild line with higher-capacity conductor | \$7,672,000   | 2% (\$133,000)        |
| Quinn – Bulverde 138 kV                           | Rebuild line with higher-capacity conductor | \$938,000     | 2% (\$19,000)         |
| Langlois – Southtown 138 kV line                  | Rebuild line with higher-capacity conductor | \$5,672,000   | 2% (\$136,000)        |
| Dane – Commonwealth Chesapeake 138 kV             | Rebuild line with higher-capacity conductor | \$16,079,000  | 3% (\$419,000)        |
| Saltspring Island – Key Largo 138 kV L            | Rebuild line with higher-capacity conductor | \$24,663,000  | 2% (\$428,000)        |
|                                                   | Total allocated to project                  |               | \$8,666,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.